

Nouraldin Farge

SOFTWARE ENGINEER - DESKTOP & LOCAL-FIRST SYSTEMS

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SUMMARY

Software engineer who has shipped three portable Windows products with reproducible release evidence and published a source-free case study for a separate private system. Owns architecture, safety boundaries, verification, and release decisions for local-first applications.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Python, SQL, HTML/CSS

Desktop & Frontend: React, Tauri, Electron, Windows, WebView2, Vite

Data & Integration: SQLite, Node.js, Express, REST, WebSockets, JSON Schema, Zod

Quality & Release: GitHub Actions, CodeQL, Vitest, pytest, Playwright, Cargo/Clippy, SBOMs, SHA-256, provenance attestations

PROJECTS

DrawScope v0.6.5 | React · Rust/Tauri · Python · SQLite

[Code](#) · [Release](#) · 2026-Present

- Built a portable research workbench backed by a reproducible 41,598-row SQLite archive spanning six lottery games, with row-level provenance and documented data gaps.
- Prevented look-ahead bias in 250 walk-forward targets by enforcing a fixed 60/40 discovery-confirmation split with contract tests across TypeScript, Python, and Rust.

GameVault v0.3.5 | React · Rust/Tauri · SQLite

[Code](#) · [Release](#) · 2026-Present

- Developed a portable game library and launcher with a relative-path SQLite catalog, launch history, backups, and optional metadata limited to allowlisted official store pages.
- Hardened review-gated ZIP imports against path traversal, unsafe Windows names, links and reparse points, and decompression abuse; required approval before promotion and supported rollback.

Day-Trading Teacher v0.32.6 | React · Rust/Tauri · local JSON · Zod

[Code](#) · [Release](#) · 2026-Present

- Created a lesson-first learning environment that connects decision plans, historical replay, paper practice, journaling, reflection, and spaced review in one local workflow.
- Implemented deterministic decimal calculations in Rust, schema-validated lesson imports, and provenance-aware review of Fidelity trade history; deliberately excluded live signals, advice, and order execution.

Research Studio alpha.21 | Electron · React · TypeScript · Express · SQLite

[Case study](#) · 2026-Present

- Architected a private Chinese/English catalog-enrichment alpha around read-only snapshots and explicit working copies, with enrichment batches capped at five records.
- Added validation for untrusted AI output, human approval gates, and recoverable exports; published a source-free case study while keeping the application, data, and build process private.

EXPERIENCE

Independent Software Engineer

2026-Present

Portfolio Products

- Owned product requirements through release for three public applications, including architecture, trust boundaries, automated verification, portable recovery, documentation, checksums, SBOMs, and provenance.
- Directed AI agents during implementation; personally reviewed changes, resolved failures, validated claims, and retained responsibility for safety, licensing, data sources, and releases.

Technical Assistant (Additional Experience)

Jun 2015-Sep 2015

Chicago Public Schools

Chicago, IL

- Supported staff and classroom technology, diagnosed hardware and software issues, and explained practical resolutions to non-technical users.

EDUCATION

Northeastern Illinois University

B.A. Computer Science, Cum Laude - GPA: 3.74

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Chicago, IL